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Studierichting: Informatica
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Bereken in $\mathbb{C}$ door gebruik van goniometrische vormen: $(-1 + i)^{10}$

$$r = \sqrt{1 + 1} = \sqrt{2}$$

$$\theta = \text{Bgtan} \left(\frac{1}{-1} \right) = \frac{3\pi}{4} + \pi = \frac{7\pi}{4}$$

$$\left(\sqrt{2} \cdot \text{cis} \frac{7\pi}{4} \right)^{10}$$

$$= (\sqrt{2})^{10} \cdot \text{cis} \left(10 \cdot \frac{7\pi}{4} \right)$$

$$= 2^5 \cdot \text{cis} \left(\frac{35\pi}{2} \right)$$

$$= 32 \cdot \text{cis} \left(\frac{3\pi}{2} \right)$$

$$= 32 \cdot \cos \left(\frac{3\pi}{2} \right) + 32 \cdot \sin \left(\frac{3\pi}{2} \right) i$$

$$= 32 \cdot 0 + 32 \cdot (-1)i$$

$$= -32i$$

References